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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 01/08/2007 |
| **PI Name:** | Midura, Ronald J Ph.D. | | |
| **Project Title:** | Effects of Simulated Weightlessness on the Repair of Lower Limb Bone Fractures and on the Number of Bone-derived Stem Cells | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Bone Loss Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 44195 | **Congressional District:** | 11 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Biomedical Research & Countermeasures 03-OBPR-04 |
| **Start Date:** | 04/01/2004 | **End Date:** | 03/31/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Cavanagh, Peter ( The Cleveland Clinic Foundation )<br>Muschler, George ( The Cleveland Clinic Foundation )<br>Vasanji, Amit ( The Cleveland Clinic Foundation ) | | |
| **Grant/Contract No.:** | NCC 9-58-BL00405 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

**Task Description:**

1) Original Aims
Aim 1: Assess the rate and extent of fracture healing in osteotomized rats in simulated weightlessness versus normal gravity. Aim 2: Assess the numbers of progenitor cell colony-forming units (CFU) and osteogenic cells per CFU in bone marrow and periosteum tissues from osteotomized rats in simulated weightlessness versus normal gravity. Aim 3: Determine the rate and extent of fracture healing in osteotomized rats in simulated weightlessness when treated with PTH or bisphosphonate therapy. Aim 4: Assess the numbers of progenitor cell CFU and osteogenic cells per CFU in bone marrow and periosteal tissues from osteotomized rats in simulated weightlessness undergoing either PTH or bisphosphonate therapy.

2) Key Findings

In year 1, as assessed by in vivo micro-computed tomography (micro-CT), a normal gravity group (weight bearing or WB) exhibited spontaneous repair of bone trauma within a 5-week healing period. PTH therapy enhanced this repair rate by increasing: 1) the rate of hard callus formation and peak volume of hard callus by ~50% over that of vehicle-control rats; and 2) the volume of cortical bone adjacent to the trauma site by ~60%. PTH therapy also increased cancellous bone content within the healing callus by ~40% over that of control rats. Mineral apposition rates were increased in PTH-treated rats as compared to control rats. PTH therapy increased the prevalence of osteogenic cell CFUs within bone marrow tissue by ~80% over that measured in vehicle-control WB rats. These findings provide support for the validity of Hypothesis 4 (PTH therapy will significantly increase the number of functional osteogenic progenitor cells in bone under simulated weightlessness).

In this second year, we quantified the extent of bone trauma repair associated with simulated weightlessness (non-weight bearing or NWB). As assessed by in vivo micro-CT, the NWB group exhibited a greatly diminished level of spontaneous repair of bone trauma as compared to the WB group within the same healing period. This diminished level was reflected in a maximum hard callus volume that was only 22% of, and a rate of hard callus formation that was only 19% of the WB group. Reacquisition of a weight-bearing status equal in time to that of the period of simulated weightlessness prior to bone trauma (NWB-WB group) did not substantially improve the bone healing response. In this case, this group exhibited a maximum hard callus volume that was only 34% of, and a rate of hard callus formation that was only 28% of the WB group.

Neither the NWB nor the NWB-WB groups exhibited measurable bending strength at the callus site by 5-weeks post-op. This finding is in stark contrast to the bending strength of the WB callus, which was measured to be 4.74 ± 1.95MPa (n=10 bones) or roughly half the bending strength of undamaged fibulae (9.57 ± 2.05MPa). Histological analysis revealed that the NWB group fibulae were reconnected by a fibrous soft tissue union, while the NWB-WB group was reconnected by a cartilaginous soft tissue union.

Our ongoing studies indicate that intermittent PTH at doses higher than those used to prevent osteoporosis can reverse the deleterious effects of simulated weightlessness on bone trauma healing. Our most recent experiments found that 120-mg/kg body weight dosage given 5 times per week to NWB rats yielded a bony union across the original osteotomy site that was roughly 50% the cancellous bone volume of the WB group over the same healing period. We are currently assessing bending strength and histology of these specimens. Altogether, these findings suggest that intermittent PTH therapy may be a candidate countermeasure against this risk of impaired bone healing in space.

3) Impact of Findings

The findings obtained within the first two years of funding support the validity of Hypothesis 1 (Native fracture healing will be adversely affected by simulated weightlessness). The implication of these findings is that bone trauma repair in astronauts on deep-space missions would likely be compromised, and presents a potential threat to mission effectiveness and astronaut health. Further, these findings also suggest that astronauts who return from extended spaceflight missions are at continued risk for impaired bone trauma repair after returning to a normal gravity environment even up to a period of time equal to that of the space mission.

Our preliminary findings suggest that the number of progenitor cell CFU is not expected to be reduced in NWB rats. Rather, the number of CFU expressing an osteogenic phenotype in bone marrow in vivo is likely reduced in NWB rats compared to WB rats. Thus, a simulated weightlessness situation may be caused by a reduction in the existing number of functional preosteoblasts and osteoblasts available for immediate use to repair bone trauma. This finding would support the validity of Hypothesis 2 (The population of functional osteogenic progenitor cells in bone will be significantly reduced by simulated weightlessness). Lastly, our results indicate that our Hypothesis 3 (PTH therapy will significantly improve fracture healing under simulated weightlessness) has merit. The implication of these findings is that PTH therapy represents a good candidate to be tested as a countermeasure to offset the deleterious effects of weightlessness on bone trauma healing.

4) Year 3 Proposed Research Plan:

Our immediate plans are to complete the currently acquired data assessments to answer definitively the goals of all of our specific aims. This includes the completion of a triplicate trial in order to meet our statistical power requirements. Further, in the upcoming year we will fully explore a potential countermeasure to offset the deleterious effects of weightlessness on bone trauma healing (PTH therapy will significantly improve fracture healing under simulated weightlessness). We are also currently drafting two manuscripts for submission to peer-reviewed publications.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Our studies found that PTH therapy increased the prevalence of osteogenic CFU within bone marrow tissue by ~80% over that measured in vehicle-control rats (normal gravity group). This finding suggests that an underlying mechanism of action of PTH in its anabolic activity for bone tissue lies at the level of increasing the pool size of functional bone cells derived from osteogenic precursor cells. These same findings suggest that PTH therapy should aid in the healing of recalcitrant bone fractures here on Earth.
Additionally, our current findings that demonstrate impaired bone healing during and after an extended exposure to simulated weightlessness should manifest relevance to the underlying causes of impaired bone healing in patients experiencing paralysis, immobility or extended bed rest. Our data indicating that intermittent PTH therapy can counteract the impairment of bone healing under a non-weight bearing situation may offer a potential treatment for augmenting bone healing in these Earth-bound, non-weight bearing patients.

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| **Task Progress:** | April 1, 2005 to March 31, 2006: In the second year of funding, we have established conclusively that long bone trauma does not heal properly in a simulated weightlessness condition (non-weight bearing or NWB). Specifically, even after 5-weeks of healing, there is only a small and incomplete hard callus bridging the bone ends in NWB rats, while over the same healing period a normal gravity condition (weight bearing or WB) exhibited a full and complete hard callus bridging. The NWB callus was comprised mostly of fibrous soft tissues and did not exhibit any measurable bending strength. In stark contrast, the WB callus was comprised mostly of woven bone and marrow tissues and exhibited roughly half the bending strength of intact bone. Overall, when these findings are extrapolated to the human setting, our results would suggest that long bone trauma to astronauts during an extended deep space mission will not heal properly. Given the increased chance that an accident could occur during a long space mission, there seems to be a realistic probability that long bone fractures will be incurred and they are likely not to heal. These findings have bearing towards answering CPR question # 2.11a: Does prolonged exposure to hypogravity lead to impaired healing of fractures?<br>A surprising finding in our investigations was the observation that a bone-healing impairment continues well after a simulated weightlessness condition is terminated and a normal weight bearing routine is re-established. Rats that were NWB for 4-weeks were returned to a WB condition for an additional 4-weeks before bone trauma was initiated. In this NWB-WB group of rats the bone trauma healing capacity was only marginally better than that exhibited by the NWB rats, and certainly not of sufficient healing quality to promote proper mechanical function (i.e., bending strength). Overall, when these findings are extrapolated to the human setting, our results would suggest that long bone trauma to astronauts that have returned back to Earth after completing an extended deep space mission will not heal properly. These findings have an additional bearing towards answering CPR question # 2.11a: Does prolonged exposure to hypogravity lead to impaired healing of fractures?<br><br>Lastly, our year 1 results and current studies using PTH in the NWB group, indicate that our third Hypothesis (PTH therapy will significantly improve fracture healing under simulated weightlessness) has merit. The implication of these findings is that PTH therapy represents a good candidate to be tested as a countermeasure to offset the deleterious effects of weightlessness on bone trauma healing. These findings have a bearing on answering CPR question #2.06: What pharmacological agents will most effectively minimize the decrease in bone mass with hypogravity? Are anabolic as well as anti-resorptive agents required? |
| **Bibliography Type:** | Description: (Last Updated: 03/01/2017) |
| **Abstracts for Journals and Proceedings** | Androjna C, Midura R J. "How Might Fracture Healing be Impaired During Spaceflight." . In Press. , Jan-2006 |
| **Abstracts for Journals and Proceedings** | Androjna C, Su X, Midura R J. "Impaired Bone Trauma Healing During Simulated Weightlessness." 2006 March. 2006 March. , Mar-2006 |
| **Abstracts for Journals and Proceedings** | Androjna, C, Su X, Midura R J. "Intermittent PTH(1-34) Enhances Bone Trauma Healing." 2005 November. 2005 November , Nov-2005 |
| **Awards** | Midura, Ronald J. "Ronald J. Midura was selected to participate on a panel of experts for the Operational and Research Musculoskeletal Summit, August 23-25, 2005 at NASA-Johnson Space Center, Houston, TX." Aug-2005 |